

V. Verification Output

(September 28, 2026)

One script re-derives every computational claim in the paper from scratch and reports pass or fail per check. Its current output follows, unedited.

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1. Realisability (thm:realis):  $c^2 = 0$  iff  $F = (s+a)(L_k + t)$ 
[PASS] k=2
[PASS] k=3
[PASS] k=4
[PASS] k=5
[PASS] k=6
[PASS] k=7
[PASS] k=8
2. Antipodal corollary:  $c^2 = 0$  identically for even  $k$ , not for odd
[PASS] k=2 (even)
[PASS] k=3 (odd)
[PASS] k=4 (even)
[PASS] k=5 (odd)
[PASS] k=6 (even)
[PASS] k=7 (odd)
[PASS] k=8 (even)
3. k=2 closed form:  $c^2 = -(s_1+s_2)(s_1+s_3)(s_2+s_3)$ 
[PASS] identity
4.  $e_2 = -k$  for every period-one symbol with  $k \geq 3$  (thm:congbound)
[PASS] k=3
[PASS] k=4
[PASS] k=5
[PASS] k=6
[PASS] k=7
[PASS] k=8
[PASS] k=9
[PASS] k=10
5. Integer certificates: the signed adjacency matrix
[PASS] signed Adj P(10,3) nullity 8 --- got 8
[PASS] signed Adj P(20,3) nullity 8 --- got 8
[PASS] signed Adj P(30,3) nullity 8 --- got 8
[PASS] signed Adj P(120,7) nullity 16 --- got 16
[PASS] signed Adj P(30,2) nullity 6 --- got 6
[PASS] signed Adj P(60,2) nullity 6 --- got 6
[PASS] Psi_7 certificate P(7,2) nullity 6 --- got 6
[PASS] Psi_7 certificate P(14,2) nullity 6 --- got 6
[PASS] Psi_7 certificate P(21,2) nullity 6 --- got 6
[PASS] Psi_7 certificate P(28,2) nullity 6 --- got 6
6. k=2 explicit certificate at the three most negative grid values
( $c^2 > 0$  is checked for all  $n$ ; the nullity is checked two ways, by
counting grid roots of the symbol exactly and, for moderate  $n$  only,
by singular values -- the three roots crowd within  $O(1/n^2)$  of  $-2$ ,
so the matrix is exact but numerically stiff for large  $n$ )
[PASS]  $c^2 > 0$  for every  $n \geq 9$ ,  $n \neq 10$  (up to 1001) --- checked 9..199, 500, 1001
[PASS] three prescribed roots are distinct and interior --- checked 9..199, 500, 1001, 5000
[PASS] P(9,2) nullity 6 numerically
[PASS] P(11,2) nullity 6 numerically
[PASS] P(20,2) nullity 6 numerically
[PASS] P(39,2) nullity 6 numerically
[PASS] P(60,2) nullity 6 numerically
7. Saved interval-certified matrices re-evaluated (k=3)
[PASS] P(17,3) nullity 8, edges nonzero --- nullity 8, min edge 0.1462
[PASS] P(18,3) nullity 8, edges nonzero --- nullity 8, min edge 0.2058

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[PASS] P(19,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0145
[PASS] P(20,3) nullity 8, edges nonzero --- nullity 8, min edge 0.1424
[PASS] P(21,3) nullity 8, edges nonzero --- nullity 8, min edge 0.3059
[PASS] P(22,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0878
[PASS] P(23,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0647
[PASS] P(24,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0405
[PASS] P(25,3) nullity 8, edges nonzero --- nullity 8, min edge 0.1584
[PASS] P(26,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0859
[PASS] P(27,3) nullity 8, edges nonzero --- nullity 8, min edge 0.2439
[PASS] P(28,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0296
[PASS] P(29,3) nullity 8, edges nonzero --- nullity 8, min edge 0.2612
[PASS] P(30,3) nullity 8, edges nonzero --- nullity 8, min edge 0.3665
[PASS] P(31,3) nullity 8, edges nonzero --- nullity 8, min edge 0.2290
[PASS] P(32,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0445
[PASS] P(33,3) nullity 8, edges nonzero --- nullity 8, min edge 0.2802
(17 saved certificates re-evaluated)
[PASS] thm:p17: a saved certificate exists for every n in [17,33] --- all 17 present
8. k=3 identity tiles and their compositions
[PASS] tiles cover a full interval [L,2L] --- lengths 21..43 (23 tiles)
[PASS] single tiles give nullity 8 --- all 23 lengths
[PASS] multi-tile concatenations give nullity 8 --- checked n up to 55
9. Certified tiles: nullity, docking exactness, coverage
[PASS] k=3: every n >= 21 is a sum of certified lengths --- 23 tiles, lengths 21..43 (= [L,2L], so L itself)
[PASS] k=3: every certified tile has nullity exactly 8 --- all 23 tiles
[PASS] k=3: docking weights exact in every certified tile --- all tiles, to 1e-13
[PASS] k=4: every n >= 29 is a sum of certified lengths --- 33 tiles, lengths 29..61 (= [L,2L], so L itself)
[PASS] k=4: every certified tile has nullity exactly 10 --- all 33 tiles
[PASS] k=4: docking weights exact in every certified tile --- all tiles, to 1e-13
[PASS] k=5: every n >= 162 is a sum of certified lengths --- 21 tiles, lengths 54..75; 44 smaller n not covered
[PASS] k=5: every certified tile has nullity exactly 12 --- all 21 tiles
[PASS] k=5: docking weights exact in every certified tile --- all tiles, to 1e-13
10. The paper's own numbers against the computed ones
[PASS] paper's certification count matches the 56 certified tiles --- all say 56
[PASS] Status section names the same k as thm:complete --- both say [2, 3, 4]
[PASS] k=3: paper's L matches the least certified tile length (21) --- L=21
[PASS] k=4: paper's L matches the least certified tile length (29) --- L=29
11. The Krawczyk test itself, re-run in EXACT arithmetic
[PASS] tile 21 at k=3: exact Krawczyk contraction --- alpha = 2.154e-10 < 1 exactly
[PASS] tile 29 at k=4: exact Krawczyk contraction --- alpha = 1.812e-10 < 1 exactly
[PASS] gn 17 at k=3: exact Krawczyk contraction --- alpha = 2.128e-11 < 1 exactly
12. Z values cross-checked against independent solvers
[PASS] Z(P(12,3)) = 7, by every method that completed --- 3/3 completed: reduced 7, unreduced 7, CP-SAT 7
[PASS] Z(P(8,2)) = 5, by every method that completed --- 3/3 completed: reduced 5, unreduced 5, CP-SAT 5
[PASS] Z(P(12,4)) = 6, by every method that completed --- 3/3 completed: reduced 6, unreduced 6, CP-SAT 6
[PASS] recorded cross-validation sweep has no disagreement --- 79 rows agree
13. The graph Riemann Hypothesis: classification of Ramanujan P(n,k)
[PASS] block decomposition reproduces the full spectrum --- max deviation 1.8e-14 over k=1..10, n<=60
[PASS] spectra match published values for Petersen and Desargues --- P(5,2) -> {3,1,-2}, P(10,3) -> {+-3,+-2,+-1}
[PASS] criterion G_k <= 7 agrees with the spectral test --- k=1..10, n<=260
[PASS] D_k(1)=-7, D_k(0)>=17, D_k(-1)=-7 (k odd) / 9 (k even), deg 2k+2 --- k=1..12; these give the bad band at
[PASS] side condition 7 + 4u T_k(u) stays positive on [-1,1] --- minimum over k=1..12 is 3.000, and |4u T_k| <=
[PASS] finiteness bound n <= 2 pi / arccos(u_k) holds for every k --- k=1..11, attained exactly at 3 of them
[PASS] parity bound n <= pi / arccos(-v_k) holds for odd k, odd n --- k=1,3,...,11, attained exactly at 4 of them
[PASS] uniform lemma: G_k > 7 on the whole interval the proof forbids --- min G = 7.004872 at the band at 0, 7.0
[PASS] uniform bound n <= pi(2+sqrt(2)) sqrt(k^2+1), halved for odd k odd n --- holds against the certified class
[PASS] every case below the bound decided in exact arithmetic --- 458 cases certified, 0 disagreements with the
[PASS] the classification is complete for k <= 10 --- 247 Ramanujan graphs; largest n per k: 1:14, 2:23, 3:32, 4:41
14. Reduction on every two-vertex cover (thm:red2)
[PASS] thm:red2: elimination gives the predicted offsets, ends nonzero --- 8 bases, incl. p>1 and w!=0 which thm

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- [PASS] thm:red2: $x \rightarrow (x, Lx)$ is a bijection onto $\ker A$, and $\text{null } A \leq 2(p+q)$ --- $\text{null } A = \text{null } R$, lift residual
- [PASS] thm:theta: $\text{null } A = \dim \ker(T-I) \leq D$ on theta covers --- 6 bases incl. four-edge; state dimension equals
- [PASS] thm:order: $\text{order} = p+q+\max(p, t_m)+\max(q, t_m)$ --- 12 bases, measured vs counted
- [PASS] thm:order: $\text{order} = D$ iff $t_m \leq \min(p, q)$ --- the characterisation holds on every base tested
15. Slice lemma (lem:slice)
- [PASS] spokes $\rightarrow +1$ and $|\text{outer}| \rightarrow 1$, nullity preserved, both parities --- max deviation $2.7e-15$
16. The corner criterion, and the classification for every k
- [PASS] both factors of D_k collapse to $Q(x, y)$, independent of k --- max deviation $4.1e-14$ over $k=1..14$
- [PASS] corner criterion agrees with the spectral test --- 1512 cases, $k=1..12$
- [PASS] lens area $3/2 - \sqrt{2} + (3/8)\log((1+\sqrt{2})/3)$ matches the integral --- 0.004321924509 of the square, i.e.
- [PASS] Dirichlet witness forces failure for every $n \geq$ threshold, every k --- threshold $n \geq 231$; worst value 0.
- [PASS] the classification is finite in BOTH variables --- 460 pairs, 324 isomorphism classes; largest $n=112$, lar
- [PASS] no case in the region is decided on a margin float64 could not see --- smallest $|\text{margin}|$ is $6.11e-05$, at
- [PASS] the all- k census reproduces the exactly certified $k \leq 10$ list --- 247 graphs at $k \leq 10$, all recovered
17. Structure of the classification: locality, gcd safety (sec:closed)
- [PASS] thm:local: Ramanujan iff no divisor $m \geq 3$ of n is bad for k --- 8280 cases against the direct criterion,
- [PASS] cor:divclosed: the Ramanujan set of each k is divisor-closed --- 202 (n, d) pairs with $d \mid n$, $d > 2k$
- [PASS] lem:mingcd: min over units of $||jc/m||$ equals $\gcd(c, m)/m$ --- $m < 120$, $c < 200$; the identity the safety th
- [PASS] thm:gcdsafe: $m/\gcd(k+1, m)$ in $\{2..7\}$ implies m is safe --- fired on 1576 of 11820 pairs, never on a bad l
- [PASS] thm:closedsmall: exact where claimed, and fails where we say it does --- holds for $k=[1, 2, 3, 4, 5, 7, 9]$
- [PASS] lem:bands: D_k has end bands only for k in $\{1, 2, 3, 4, 5, 7, 9\}$ (exact Sturm root isolation, $k \leq 45$) --- end ba
- [PASS] lem:bands: D_k is an even polynomial exactly for odd k --- $k \leq 45$
18. The Riemann Hypothesis on every cubic cyclic cover (thm:covcorner)
- [PASS] character blocks reproduce the spectrum on both two-vertex bases --- max deviation $8.7e-15$
- [PASS] the two-loop spectrum does not depend on the edge voltage c --- max deviation $3.3e-15$ over every c
- [PASS] corner criterion decides every two-loop cover --- 531 covers, brute-forced from the full adjacency matrix
- [PASS] theta criterion decides every cyclic Haar cover --- 384 covers, brute-forced from the full adjacency matr
- [PASS] both uniform bounds hold on every Ramanujan cover found --- 485 Ramanujan covers, none above its bound
- [PASS] no cover of ANY fixed cubic base is Ramanujan past its threshold --- $\text{theta}(0, 1, 2)$: max $n=15 < 156$; $\text{theta}(a$
19. The cover ceiling: two theorems and a refutation
- [PASS] thm:redpath: degree span of $\det M(\zeta)$ is $2 \sum(p_t)$ --- 8 path-with-loops bases, $m = 2$ to 4 , checked sy
- [PASS] thm:redpath: non-equivariant weights cannot beat $2 \sum(p_t)$ --- $p=(1, 2)$ $n=9$ $D=6$; $p=(1, 3)$ $n=11$ $D=8$; $p=(2$
- [PASS] thm:redcycle: degree span is $2|W|$ with W the holonomy --- 7 cycle bases, checked symbolically
- [PASS] thm:redcycle: the cover is 2-regular with $\gcd(|W|, n)$ components --- 42 (base, n) pairs -- this is why $M(C$
- [PASS] thm:cexcover: star-with-a-loop breaks the ceiling, $\text{null } A = n$ vs $D = 2$ --- 9 (n, weight) pairs, ranks com
- [PASS] rem:mindeg: minimum degree 2 does not repair it $(K_{\{2,3\}}, D=4)$ --- $n=5$: $\text{null}=6$ vs $D=4$; $n=7$: $\text{null}=8$ vs $D=4$
- [PASS] the diagonal-zeroing attack alone never exceeds D on a regular base --- 10 (regular base, n) pairs, exact
- [PASS] thm:k4rankone: rank-one triangle blocks give $\text{null } A = n$ on the K_4 cover, against $D = 6$ (regular-base ceiling)
- [PASS] thm:cover needs $\det M(\zeta) \neq 0$: the rank-one equivariant matrix has $\det M(\zeta) = 0$ identically --- ev
- [PASS] conj:leading: top coefficient of $\det M(\zeta)$ is a monomial in edge weights on every base where the ceiling
- [PASS] conj:leading: top coefficient contains a free diagonal on every base where the ceiling fails --- star-4 +
20. The independent-set obstruction, and the cubic gap family
- [PASS] cor:regblocks: d -regular implies $|N(S)| \geq |S|$ for independent S --- 2798 (regular graph, independent set
- [PASS] sec:gapfamily: the K_4 covers are cubic, connected, with GENERIC $D = 6$ for all n --- $n = 4..10$, $|V| = 16..$
- [PASS] prop:k4cert: the integer weights realise $16(z^7-1)/(z-1)$ --- coefficients of $z^3 \det M$ all equal 16
- [PASS] prop:k4cert: exact nullity over Z equals the degree span --- $n=7$: $\text{null}=6=D$; $n=14$: $\text{null}=6=D$
- [PASS] prop:k4z: $Z = n+2$ by CP-SAT with optimality proved --- $n=4$: $Z=6$; $n=5$: $Z=7$; $n=6$: $Z=8$; $n=7$: $Z=9$; $n=8$: $Z=10$
21. The deficit (prop:deficit)
- [PASS] $\delta(n, k) < 3 - 2\sqrt{2}$ for every n , and δ^* rises towards it --- max $\delta^* = 0.129473 < \tau = 0.171$
22. $M(P(10, 2))$ exactly, and the period-one ceiling
- [PASS] $M(P(10, 2))=6$: a period-two solution over Q exists --- $a=1/5$, $c=1$, $e=1$, $f=-5$, $s=-1$, $t=-15$
- [PASS] $M(P(10, 2))=6$: the witness carries exactly the $P(10, 2)$ pattern --- 20×20 , symmetric=True, missing=0, extra
- [PASS] $M(P(10, 2))=6$: exact nullity over Q equals 6 --- exact rank 14, nullity 6; with $M \leq Z = 6$ this gives $M(P($
- [PASS] no period-one matrix attains nullity 6 at $P(10, 2)$ --- 10 block-class subsets of total dimension 6, 0 admi
- [PASS] period-one ceiling explains both missing certificates --- $P(10, 2)$: bound $5 < 6$; $P(24, 4)$: bound $8 < 10$ --
- [PASS] CONTROL: the period-one bound contradicts no existing certificate --- 16 certified (n, k) checked, 0 viola
- [PASS] the period-one bound reproduces the paper's two exceptions at $k=2$ --- blocked n at $k=2$: $[8, 10]$ -- exactl
23. Emit the manuscript's numbers from the certificates on disk
23. Classification of rational-symbol rotation-invariant certificates (thm:ratclass)

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[PASS] thm:ratclass: the admissible classes are 7,9,15,20 | 10,24,42 | 60,70,90 | 24,126 | none | 120 for k=2..7
[PASS] thm:ratclass: every listed certificate has nullity 2k+2 on its class and 0 off it --- 26 certificates, k=
[PASS] manuscript numbers emitted from the certificates on disk --- 56 tiles at k=3,4 + 21 at k=5; L(3)=21, L(4)
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ALL CHECKS PASSED
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145 checks in 332.6 s (5.5 min) on 10 cores
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SUITE_DONE
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VI. A Certification Transcript

(September 27, 2026)

What “interval-certified” means in practice, for one family of tiles. Each entry is an independent Krawczyk contraction proof with its own unknown count, matrix norm and residual; nothing is shared between them.

certifying 22 identity tiles at $k=3$

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1=22: 398 unknowns, 352 equations, 324 structural, 28 dependent; 40 docking pins, 74 fixed total
      structural 3.11e-61, dependent 2.72e-61, ||Y||=5.39e+02, alpha=6.70e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=23: 419 unknowns, 368 equations, 340 structural, 28 dependent; 40 docking pins, 79 fixed total
      structural 3.11e-61, dependent 4.67e-61, ||Y||=7.55e+02, alpha=9.25e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=24: 440 unknowns, 384 equations, 356 structural, 28 dependent; 40 docking pins, 84 fixed total
      structural 6.22e-61, dependent 1.94e-61, ||Y||=2.48e+03, alpha=3.24e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=25: 461 unknowns, 400 equations, 372 structural, 28 dependent; 40 docking pins, 89 fixed total
      structural 6.22e-61, dependent 2.24e-61, ||Y||=9.96e+02, alpha=1.23e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=26: 482 unknowns, 416 equations, 388 structural, 28 dependent; 40 docking pins, 94 fixed total
      structural 3.5e-61, dependent 1.94e-61, ||Y||=5.16e+02, alpha=6.88e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=27: 503 unknowns, 432 equations, 404 structural, 28 dependent; 40 docking pins, 99 fixed total
      structural 6.22e-61, dependent 1.56e-61, ||Y||=1.51e+03, alpha=1.89e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=28: 524 unknowns, 448 equations, 420 structural, 28 dependent; 40 docking pins, 104 fixed total
      structural 6.22e-61, dependent 2.48e-61, ||Y||=1.04e+03, alpha=1.31e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=29: 545 unknowns, 464 equations, 436 structural, 28 dependent; 40 docking pins, 109 fixed total
      structural 3.31e-61, dependent 1.56e-61, ||Y||=3.84e+02, alpha=4.77e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=30: 566 unknowns, 480 equations, 452 structural, 28 dependent; 40 docking pins, 114 fixed total
      structural 6.22e-61, dependent 3.69e-61, ||Y||=5.86e+02, alpha=7.43e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=31: 587 unknowns, 496 equations, 468 structural, 28 dependent; 40 docking pins, 119 fixed total
      structural 4.67e-61, dependent 6.51e-61, ||Y||=6.67e+02, alpha=8.49e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=32: 608 unknowns, 512 equations, 484 structural, 28 dependent; 40 docking pins, 124 fixed total
      structural 3.11e-61, dependent 2.14e-61, ||Y||=6.86e+02, alpha=8.71e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=33: 629 unknowns, 528 equations, 500 structural, 28 dependent; 40 docking pins, 129 fixed total
      structural 2.53e-61, dependent 2.33e-61, ||Y||=5.64e+02, alpha=7.14e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=34: 650 unknowns, 544 equations, 516 structural, 28 dependent; 40 docking pins, 134 fixed total
      structural 4.76e-61, dependent 4.08e-61, ||Y||=2.88e+02, alpha=3.66e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=35: 671 unknowns, 560 equations, 532 structural, 28 dependent; 40 docking pins, 139 fixed total
      structural 3.11e-61, dependent 5.25e-61, ||Y||=1.43e+03, alpha=1.77e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=36: 692 unknowns, 576 equations, 548 structural, 28 dependent; 40 docking pins, 144 fixed total
      structural 7.0e-61, dependent 3.11e-61, ||Y||=2.17e+03, alpha=2.79e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=37: 713 unknowns, 592 equations, 564 structural, 28 dependent; 40 docking pins, 149 fixed total
      structural 3.11e-61, dependent 3.4e-61, ||Y||=8.29e+02, alpha=1.06e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=38: 734 unknowns, 608 equations, 580 structural, 28 dependent; 40 docking pins, 154 fixed total
      structural 3.11e-61, dependent 2.14e-61, ||Y||=8.16e+02, alpha=1.01e-10, nullity 8, docking exact True, min|edge| 0.1502 CEF
1=39: 755 unknowns, 624 equations, 596 structural, 28 dependent; 40 docking pins, 159 fixed total
      structural 3.11e-61, dependent 4.67e-61, ||Y||=5.92e+02, alpha=7.70e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=40: 776 unknowns, 640 equations, 612 structural, 28 dependent; 40 docking pins, 164 fixed total
      structural 3.69e-61, dependent 1.94e-61, ||Y||=1.15e+03, alpha=1.43e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=41: 797 unknowns, 656 equations, 628 structural, 28 dependent; 40 docking pins, 169 fixed total
      structural 6.22e-61, dependent 5.45e-61, ||Y||=8.81e+02, alpha=1.11e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=42: 818 unknowns, 672 equations, 644 structural, 28 dependent; 40 docking pins, 174 fixed total
      structural 6.22e-61, dependent 3.11e-61, ||Y||=1.65e+03, alpha=2.12e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=43: 839 unknowns, 688 equations, 660 structural, 28 dependent; 40 docking pins, 179 fixed total

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